

Research Paper

Experimental performance evaluation of air-based photovoltaic–thermal collector with rectangular turbulators and longitudinal fins

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ABSTRACT

An air-based photovoltaic–thermal collector (PVTC) simultaneously generates electricity and heats air using energy from incident solar radiation. Its simple design ensures low initial costs and easier maintenance. However, its thermal efficiency is limited by the low heat conductivity of air. To address this shortcoming, considerable research has focused on enhancing either the heat transfer area or the heat transfer coefficient. However, only one of these two methods has typically been utilized, and information regarding PVTCs that incorporate both technologies is lacking. Thus, a novel air-based PVTC featuring longitudinal fins and rectangular turbulators is proposed herein. In this PVTC, the fins enlarge the heat transfer area, while the turbulators improve the heat transfer coefficient by creating turbulence near the fin surface. The proposed PVTC (PVTC-LFRT) was experimentally evaluated under real-world weather conditions and also compared with two other PVTC configurations: one with a smooth air channel (PVTC-SMAC) and one with only fins (PVTC-LF) to reveal the potential thermal and electrical gains achievable with the proposed design. As a result, PVTC-LFRT achieved an electrical efficiency of 17.06 %, whereas PVTC-SMAC and PVTC-LF recorded corresponding values of 16.55 % and 16.8 %, respectively. Moreover, PVTC-LFRT exhibited a thermal efficiency of 38.93 %, while PVTC-SMAC and PVTC-LF achieved thermal efficiencies of 22.68 % and 33.56 %, respectively. Finally, the total daily energy output of PVTC-LFRT from incident solar radiation was 42.73 % and 11.2 % higher than those of PVTC-SMAC and PVTC-LF, respectively. These findings verify that incorporating fins with turbulators into a PVTC effectively enhances its electrical and thermal efficiency. Consequently, the feasibility of the proposed system could be confirmed, and its promotion is considered worthwhile.

1. Introduction

With the continuous increase in energy demands and the use of fossil fuels worldwide, the associated environmental issues are becoming progressively more important from a global perspective. To resolve these environmental problems, renewable energy systems are being increasingly adopted. Among such systems, photovoltaic (PV) power generation is the most commonly utilized option, capable of producing electricity directly from incident solar radiation while incurring a low initial cost. However, PV panels typically convert only 10–20 % of the incident solar intensity into useful electricity (Bayrak et al., 2019), and the remaining energy is dissipated in the form of waste heat. In addition,

as the temperature of a PV cell rises during power generation, its performance diminishes.

To overcome such problems, Wolf (1976) introduced the idea of a photovoltaic–thermal collector (PVTC). It is a system in which the PV cells are cooled using a thermal fluid or coolant, whose temperature is lower than that of the PV cells. This prevents the electrical efficiency of the PV panel from being reduced under high operating temperatures. In addition, the coolant that receives waste heat from the PV panel can be used to heat residential spaces, water, etc. Consequently, this system can concurrently generate electricity and heat from solar energy with a higher electrical efficiency. Typically, PVTCs are categorized as air-based or liquid-based, depending on the coolant. In air-based PVTCs, the PV panels are cooled by air. Such PVTCs incur lower initial

Abbreviation: PVTC, photovoltaic–thermal collector; PVTC-SMAC, PVTC with smooth air channel; PVTC-LF, PVTC with only longitudinal fins; PVTC-LFRT, PVTC with longitudinal fins and rectangular turbulators.

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Nomenclature		$\dot{w}$	Power generation (W/m ²)
<i>Symbols</i>		<i>Greek letters</i>	
A	Area (m ²)	η	Efficiency (-)
C_p	Specific heat (J/kgK)	<i>Subscripts</i>	
C_{te}	Thermal equivalent factor (-)	air	Air
G	Solar intensity (W/m ²)	e	Electrical
h	Height (mm)	f	Longitudinal fin
I	Current (A)	in	Inlet
l	Length (mm)	out	Outlet
$\dot{m}$	Mass flow rate (kg/s)	pv	Photovoltaic panel
p	Pitch (mm)	pvtc	Photovoltaic–thermal collector
$\dot{q}_{air}$	Heat gain of air (W/m ²)	t	Turbulator
t	Thickness (mm)	te	Thermal equivalent
T	Temperature (°C)	th	Thermal
V	Voltage (V)		
w	Width (mm)		

investment costs and are easier to maintain than liquid-based PVTCs, as there are no concerns about leaking or freezing of the coolant. However, their thermal efficiency is lower owing to the low heat capacity and thermal conductivity of air compared with liquid coolants (Hussain et al., 2013). To address this drawback, numerous studies have focused on enhancing the performance of air-based PVTCs.

1.1. Literature review

To date, considerable research has been conducted on thermal performance enhancement of an air-based PVTC. A common method to enhance the thermal performance of a PVTC is either to enlarge the heat transfer area or to improve heat transfer coefficient. Generally, fins have been employed to extend the heat transfer area between the PV panel and the airflow within the air channels in an air-based PVTC. Othman et al. (2007) developed a PVTC with a double-path air channel and installed fins parallel to the airflow direction. They observed that the fins and double channels effectively increased the thermal efficiency of the PVTC. Kumar and Rosen (2011) installed vertical fins beneath the PV panel of a PVTC and performed numerical analysis to evaluate the performance of the PVTC, both with and without the fins. Their findings revealed that the fins reduced the PV cell temperature by 15 °C and increased the thermal efficiency by 10 %. Mojumder et al. (2016) developed an analytical model of a PVTC with longitudinal fins and evaluated its performance with various fin numbers and air mass flow rates. Their results presented that the increasing fin numbers and air mass flow rate provided higher thermal and electrical efficiencies. Additionally, the proposed PVTC achieved a maximum thermal efficiency of 56.19 % and an electrical efficiency of 13.75 % under the investigated conditions. Bayrak et al. (2019) attached straight fins to the PV panel of a PVTC and investigated the variations in its cooling performance with various fin parameters under natural convection conditions. Based on their observations, fins with appropriate parameters could enhance the electrical efficiency of the PV panel by up to 11.55 %. Kalkan et al. (2019) used computational fluid dynamics (CFD) to investigate the effects of ambient temperature, air velocity, flow channel length, and flow channel height on the performance of a PVTC with fins. They reported that the overall efficiency depended predominantly on the flow channel height and air velocity and that attaching fins could improve the overall efficiency by up to 19 %. Arslan et al. (2020) experimentally evaluated the effectiveness of a PVTC with an L-shaped copper fin attached behind the PV panel. Their results showed that such fins could reduce the temperature of the PV panel, as well as increase the electrical efficiency by 0.42 %. Further, they recorded electrical and thermal efficiencies of 13.98 % and 49.5 % during their experiments.

Elbreki et al. (2021) compared the passive cooling performance of two different PV panels, one with lapping fins and the other with longitudinal fins. Their findings revealed that using lapping fins can improve the passive cooling performance of the PV panel. Shoeibi et al. (2021) analyzed the performance of a PVTC with a compound parabolic concentrator and V-shaped fins. They observed that a PVTC equipped with fins performs better than one without fins and that a larger number of fins provides better performance. Huang et al. (2021) evaluated the energy and exergy performance of a PVTC with fins and compared its performance with those of traditional solar air collector. This research reported that the PVTC with fins achieved a higher exergy efficiency of about 12.22 %, compared to 8.71 % for a solar air collector. Shrivastava et al. (2022) assessed the energy and exergy performance of a PVTC equipped with a fin and baffle in the air duct. Their research revealed that utilizing the fin and baffle increased both the energy and exergy efficiency of the PVTC. Agrawal et al. (2022) estimated the annual electrical and thermal output of a PVTC integrated with rectangular fins using a mathematical model. They reported that integrating fins with the PVTC resulted in 0.2 % higher electrical output and 3 % higher thermal output compared to conventional PVTC. Kong et al. (2022) proposed and evaluated a solar dryer system using a PVTC coupled with longitudinal fins and presented that the PVTC yielded an energy efficiency of 61.4 % and an exergy efficiency of 9.95 %. Shafiee et al. (2023) attached a fin behind the PV panel and used a fan to cool the PV panel more effectively. They confirmed that the PV panel's temperature was reduced effectively by the fin, which enhanced the electrical efficiency. Zhao et al. (2023) compared the performance of a PVTC with fins to that of a PVTC without fins and reported the PVTC with fins achieved a total efficiency of 61.08 %, whereas the PVTC without fins showed 26.27 %. Alshibil et al. (2023) proposed a PVTC combined with lover-shaped fins and copper tubes to heat both air and water. This research reported that the electrical and thermal efficiencies were 6.7 % and 34.26 % when only air was heated in the proposed PVTC. Cetina--Quiñones et al. (2023) evaluated four different PVTC configurations: no fins, lineal fins, zig-zag fins, and wavy fins, and revealed the wavy fins provided the highest energy efficiency. Abdallah et al. (2024) performed a parametric study of a PVTC combined with vertically installed fins and reported that the optimal design of the fins can lower the average PV panel temperature by 6.2 % compared to a PVTC with fins without optimization. The approach of enlarging the heat transfer surface has also been used for solar air heaters (SAHs), which typically consist of two parallel flat plates, similar to air-based PVTCs. One side of each plate is thermally insulated, while the other side allows heat flux. Various types of fins have been used to enhance the performance of SAHs, including offset fins (Rai et al., 2018), staggered C-shaped fins

(Saravanan et al., 2021), and louvered fins (Chand et al., 2022). More information about the fins used to extend the heat transfer area in air-based solar collectors can be found in Ref. (Kabeel et al., 2017; Singh et al., 2022).

The other method to enhance the air heating performance of an air-based PVTC is improving the heat transfer coefficient between the heated surface and the flowing air. This can be achieved by attaching turbulators to the heated surface or installing baffles in the air channel (Varun Kumar et al., 2018). Jha et al. (2019) proposed a PVTC with a wavy absorber plate at the bottom of the air channel and compared its performance with a PVTC using a flat absorber plate. In their research, it was demonstrated that the wavy absorber plate provided 16 % and 1.2 % higher thermal and net electrical energy than PVTC utilized a flat plate absorber. Jha et al. (2020) compared the PVTC inserting a wavy plate with traditional PVTC. Their results indicated that the proposed PVTC achieved 3.5 % higher total exergy efficiency than that of traditional PVTC. Song et al. (2023) installed hole plates and baffles in the air duct of a PVTC to improve thermal efficiency. Using a mathematical model, the thermal performance and friction loss were evaluated with various opening ratios of holes in the plates. They reported that the highest thermal efficiency was achieved when the opening ratio was 0.4 and that the average thermal efficiency ranged from 0.3 to 0.58. Choi and Choi (2023) installed a transverse rib with a nonuniform cross-section in an air-based PVTC integrated with a single-pass double-flow air duct and evaluated its energy conversion performance through numerical analysis. Their research presented that installing the ribs could enhance the PVTC's daily electrical and thermal energy output. Xie et al. (2024) assessed an air-cooled PVTC with a double serpentine runner to improve heat transfer from the PV panel to the air and evaluated its performance through CFD analysis, considering pressure drop. Their results revealed that their PVTC could provide similar thermal and electrical performance to previously studied PVTCs with low-pressure drop. Dehghan et al. (2024) integrated a converging air channel with a PVTC to enhance heat transfer in the air channel. This study reported that the converging air channel with a middle inlet and side outlets performed better than the other converging air channel with a different size inlet and outlets. Moreover, they conducted an economic analysis and concluded that the PVTC with a converging air channel is more economical than traditional PV panels. This technique of increasing the heat transfer coefficient has been utilized more widely in SAHs than in PVTCs. Yadav and Bhagoria carried out CFD analyses on an SAH featuring circular (Yadav and Bhagoria, 2013), square (Yadav and Bhagoria, 2014a), and triangular ribs (Yadav and Bhagoria, 2014b) in the air channel. Their findings showed that these ribs could effectively enhance heat transfer from the heated plate to the flowing air. To improve the heat transfer in the air channel of an air-based solar collector, various types of turbulators have been studied in addition to ribs; these include multi-V perforated baffles (Kumar and Kim, 2016), forward-facing chamfered rectangular ribs (Kumar et al., 2018), T-shaped obstacles (Ahn and Kim, 2020), protrusion ribs (Alam et al., 2021), and so on (Chhapparwal et al., 2019; Nidhul et al., 2021). In addition, a more comprehensive review related to the various types of air-based PVTC can be found in recent review papers (Tiwari et al., 2023; Oh et al., 2023; Herrando et al., 2023; Jha et al., 2023; Meria et al., 2024).

As evident from the literature review above, the heat transfer between the PV panel and the flowing air in an air-based PVTC can be enhanced by enlarging the heat transfer area or improving the heat transfer coefficient. However, with PVTCs, previous research has primarily focused on the former approach rather than the latter. Moreover, only one of these two methods has typically been adopted in PVTCs. Combining the two techniques by installing turbulators on the fin surface can not only enlarge the heat transfer area but also increase the heat transfer coefficient at the fin surface. However, to the best of our knowledge, no study has yet incorporated both turbulators and fins into an air-based PVTC, and the potential gains in thermal and electrical efficiency achievable with this combined approach have not yet been

adequately revealed. This research gap motivates us to conduct this study.

1.2. Scope and aim of the work

This study proposes a novel air-based PVTC featuring a rectangular turbulator and longitudinal fins to incorporate both heat transfer area enhancement and heat transfer coefficient increase technologies. The longitudinal fins are installed beneath the PV panel, with the rectangular turbulators attached to the fin surface. The installed fins extend the heat transfer area. Meanwhile, the rectangular turbulators enhance the heat transfer coefficient at the fin surface by creating turbulence near the fin surface, promoting the mixing of heated air close to the fin surface with cooler air farther away. Thus, in this suggested design, the heat transfer area in the air channel of the PVTC can be enlarged, while the heat transfer coefficient at the fin surface is improved, which enhances the thermal performance of the PVTC.

In the present work, the proposed PVTC was fabricated, and its thermal and electrical efficiencies were evaluated under the actual weather conditions. In addition, a PVTC with a smooth air channel (PVTC-SMAC) and another with only longitudinal fins (PVTC-LF) were also assessed under the same weather conditions on the same day. The objectives of this study are: (a) to evaluate the thermal and electrical performance of the proposed PVTC, (b) to confirm the additional thermal and electrical energy harvested by the proposed PVTC compared to a PVTC with a smooth air channel and a PVTC with only longitudinal fins, and (c) to verify the feasibility of its design.

2. Experiment apparatus and method

2.1. Details of air-based PVTC

In this subsection, the PVTCs that have been experimentally studied are explained in detail. Fig. 1 depicts a schematic of the proposed PVTC, which was composed of a conventional PV panel, an air channel, a longitudinal fin, and a rectangular turbulator installed on the fin surface. The air channel was fabricated using an aluminum duct, with insulators attached to its side and bottom surfaces. The specifications of the PV panel (Q.PEAK BFR-G4.4 310 W) are detailed in Table 1.

Fig. 2 presents the front and side views of the proposed PVTC. As depicted in the figure, vertically flipped L-shaped fins are affixed to the back surface of the PV panel, and rectangular turbulators are installed on the fin surface. The total height (h_{pvtc}), width (w_{pvtc}), and length (l_{pvtc}) of the PVTC are 80 mm, 1000 mm, and 1670 mm, respectively. The side and the bottom surfaces of the air channel in the PVTC are insulated with 5 mm and 40 mm insulators, respectively. The height (h_f), width (w_f), and thickness of the longitudinal fin are 40 mm, 40 mm, and 1 mm, respectively. Within the air channel, a total of 21 longitudinal fins are installed, thus creating 22 sub-air channels. Each rectangular turbulator is 39 mm in height (h_t), 39 mm in width (w_t), 8.52 mm in thickness (t_t), and 8.52 mm in length (l_t). Seven rectangular turbulators with a pitch (p_t) of 223.08 mm are installed in each channel created by the longitudinal fins. Therefore, a total of 152 turbulators are installed in the entire PVTC. Table 2 summarizes all the PVTC dimensions.

As baselines for comparing the proposed design, two other PVTCs were evaluated in this study: one with a smooth air channel (PVTC-SMAC) and the other with only longitudinal fins (PVTC-LF). Fig. 3 shows the actual view of the rear surface of the PV panel used in the three PVTCs. PVTC-SMAC did not feature any heat transfer enhancement equipment within the PVTC, as seen in Fig. 3(a), while PVTC-LF was equipped solely with longitudinal fins, as observed in Fig. 3(b). Fig. 3(c) shows the actual view of the PVTC combined with both longitudinal fins and rectangular turbulators, designated as PVTC-LFRT in this study.

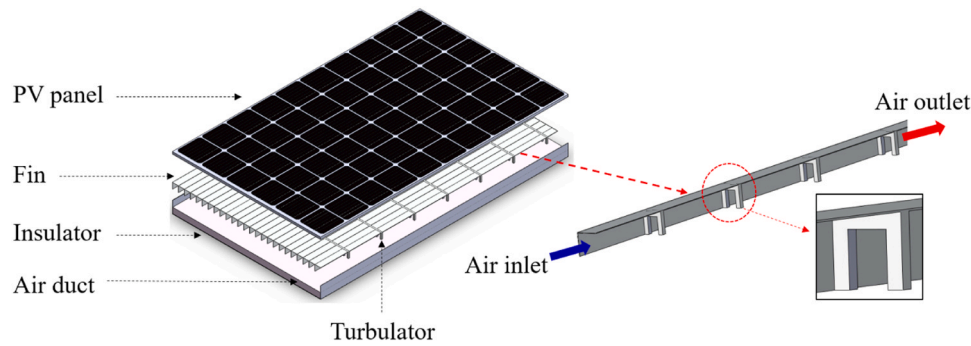


Fig. 1. Schematic of proposed PVTC coupled with fins and turbulators.

Table 1

Specifications of PV panel used in this study.

Parameter	Value
Model	Q.PEAK BFR-G4.4 310
Panel size (mm)	1670 × 1000 × 32
Standard electrical efficiency (%)	18.6
Temperature coefficient (%/K)	−0.39
Maximum voltage output (V)	32.75
Maximum current output (A)	9.46
Maximum power output (W)	310

2.2. Experimental method

This subsection presents experimental methods in detail. Fig. 4 illustrates an actual experimental apparatus of the three different air-based PVTCs. The PVTCs were installed on the rooftop of Building 2, Pukyong National University, Busan, Republic of Korea (latitude: 35°6.98N, longitude: 129°5.39E). The installation angle of PVTCs was 22°. The three PVTCs were assessed under the same weather conditions on the same day and at the same air flow rate of 0.088 kg/s. The experiments were carried out in clear weather from 10:00 am to 15:00. Table 3 summarizes the experimental conditions.

Fig. 5 shows a schematic of the experimental apparatus, including the locations of the measuring points. The ambient air flows directly into the air channel of the PVTC and is heated as it retrieves waste heat from

the PV panel. Simultaneously, the PV panel generates electrical energy from the incident solar radiation. During the experiment, various parameters were measured to evaluate the performance of the PVTCs, including solar intensity, ambient air temperature, air temperature at the PVTC outlet, and the voltage and current produced by the PVTCs. The solar intensity was measured using a pyranometer installed parallel to the PVTCs. Both the ambient and outlet air temperatures were measured using a T-type thermocouple. Voltage and current were measured using a voltmeter and an amperemeter, respectively. The air velocity was measured using an anemometer at the outlet of the air ducts of the PVTCs. All the measured data were recorded by the multifunction data logger. The specifications of the measuring devices are listed in Table 4.

2.3. Performance indices

This study evaluated the PVTC's performance based on thermal efficiency, electrical efficiency, and thermal equivalent efficiency. The thermal efficiency of the PVTC is defined as the ratio of the useful thermal energy achieved from the PVTC to the incident solar intensity. It can be written as (Shoeibi et al., 2021):

$$\eta_{th} = \frac{\dot{q}_{air}}{G} \quad (1)$$

where $\dot{q}_{air}$ and G denote the heat gain of the air flowing within the PVTC

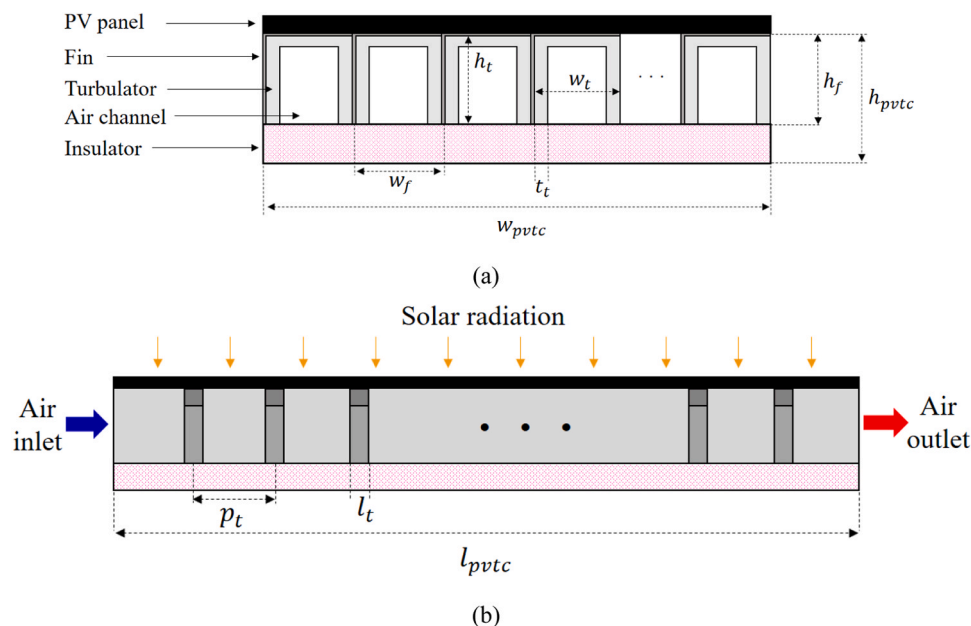


Fig. 2. (a) Front and (b) side views of proposed PVTC coupled with fins and turbulators.

Table 2
Dimensions of proposed PVTC.

Parameter		Value
PVTC	Height, h_{pvtc} (mm)	80
	Width, w_{pvtc} (mm)	1000
	Length, l_{pvtc} (mm)	1670
	Area, A_{pvtc} (m ²)	1.67
Longitudinal fin	Height, h_f (mm)	40
	Width, w_f (mm)	40
Rectangular turbulator	Height, h_t (mm)	39
	Width, w_t (mm)	39
	Length, l_t (mm)	8.52
	Thickness, t_t (mm)	8.52
	Pitch, p_t (mm)	223.08

and the incident solar intensity, respectively. The useful heat gain of the PVTC is defined as follows:

$$\dot{Q}_{air} = \frac{\dot{m}_{air} C_{p,air}}{A_{pvtc}} (T_{air,out} - T_{air,in}) \quad (2)$$

where $\dot{m}_{air}$, $C_{p,air}$, $T_{air,out}$, and $T_{air,in}$ are air mass flow rate, the specific heat of air, outlet air temperature, inlet air temperature, and the area of the PVTC, respectively. The power generation of the PVTC can be attained by:

$$\dot{W}_{pvtc} = \frac{V_{pv} I_{pv}}{A_{pvtc}} \quad (3)$$

where V_{pv} and I_{pv} are the voltage and current generated by the PV panel. The electrical efficiency of the PVTC is determined using the following relation:

$$\eta_e = \frac{\dot{W}_{pvtc}}{G} \quad (4)$$

Generally, electrical energy is deemed to have superior quality compared with thermal energy. Thus, many studies have evaluated the total energy conversion efficiency of PVTCs in terms of thermal equivalent efficiency, which incorporates the thermal equivalent factor (C_{te}). Thermal equivalent efficiency is defined as (Fudholi et al., 2014; Sardarabadi et al., 2014; Aste et al., 2015; Slimani et al., 2017)

$$\eta_{te} = \frac{\eta_e}{C_{te}} + \eta_{th} \quad (5)$$

In this study, C_{te} was set as 0.38, which is a commonly adapted value in previous studies (Fudholi et al., 2014; Sardarabadi et al., 2014; Slimani et al., 2017).

2.4. Uncertainty analysis

The experimental results relevant to the PVTCs could contain un-

certainities because the sensors used for the various measurements themselves have inherent errors. Thus, to correctly assess the performance indices of the PVTCs, detecting the errors correlated with performance indices is essential (Zhang et al., 2014). The uncertainty for each parameter is calculated as follows:

$$\delta y = \sqrt{\left(\frac{\delta y}{\delta x_1} \delta x_1\right)^2 + \left(\frac{\delta y}{\delta x_2} \delta x_2\right)^2 + \dots + \left(\frac{\delta y}{\delta x_n} \delta x_n\right)^2} \quad (7)$$

where δx_1 , δx_2 , ..., δx_n are the uncertainties associated with the directly measured values. The main parameters used for evaluating PVTC performance in this study are $\dot{Q}_{air}$, η_{th} , $\dot{W}_{pvtc}$, η_e , and η_{te} ; their respective uncertainties are listed in Table 5.

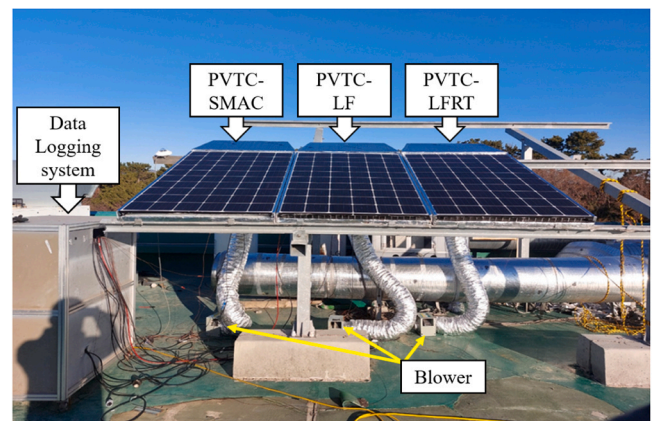


Fig. 4. Actual view of the experimental apparatus.

Table 3
Experimental conditions.

Parameter	Value
Location	Building 2, Pukyong National University, Busan, Republic of Korea
Date (dd-mm-yy)	03-01-23
Operating time (hh:mm)	10:00–15:00
Air mass flow rate (kg/s)	0.088
PVTC installation angle (°)	22

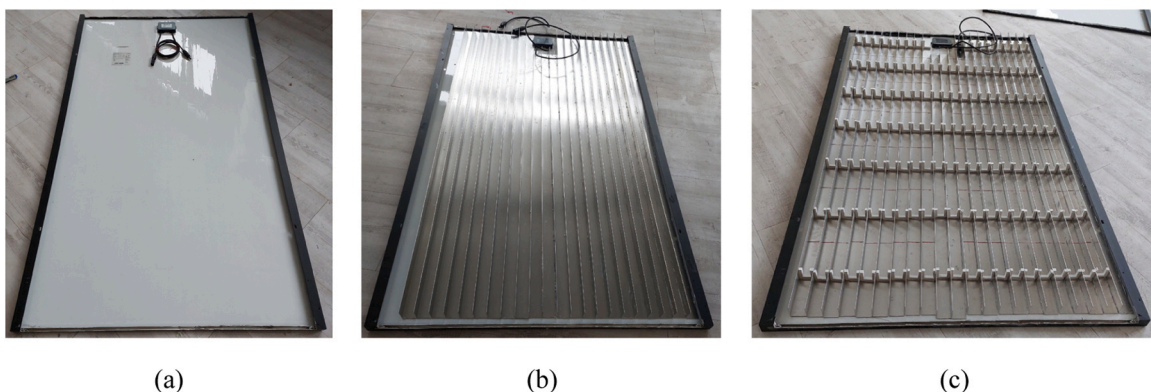


Fig. 3. Actual view of the rear surface of PV panel used in three different PVTCs: (a) PVTC-SMAC, (b) PVTC-LF, (c) PVTC-LFRT.

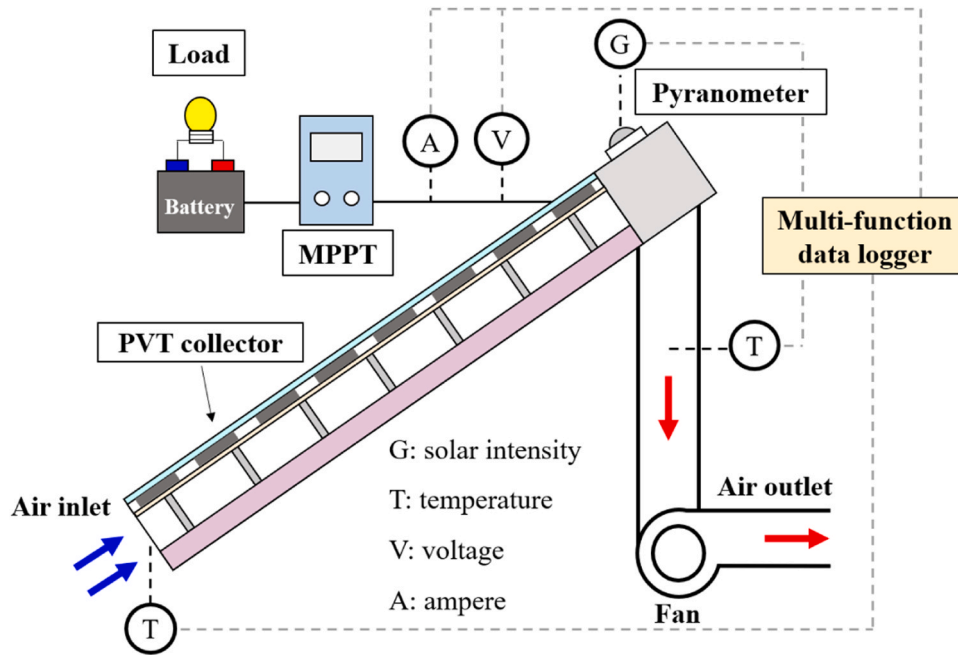


Fig. 5. Schematic of experimental apparatus with the locations of measuring points.

Table 4
Specification of measuring devices.

Measuring device (Model)	Range	Uncertainty
Pyranometer (MS–802)	0–4000 W/m ²	±2 %
Thermocouple (T-type)	–281 to 370 °C	±0.75 %
Voltmeter (MT4Y-DV–43)	0–50 V	±0.56 %
Amperemeter (MT4Y-DA–43)	0–5 A	±0.56 %
Anemometer (Kanomax 6531–2 G)	0.01–9.99 m/s	±0.015 %
Multifunction data logger (Agilent 34972 A)	–	–

Table 5
Uncertainties of parameters for each experimental condition.

Parameter	Uncertainty
$\dot{q}_{air}$	±1.06 %
η_{th}	±2.26 %
$\dot{W}_{pvtc}$	±0.79 %
η_e	±2.15 %
η_{te}	±2.40 %

3. Results and discussion

3.1. Weather conditions

In this subsection, the weather data recorded during the experiment are presented. Fig. 6 shows the solar intensity and ambient temperature throughout the experiment. The ambient temperature varied between 2.39 °C and 6.61 °C, while the solar intensity ranged from 556.02 W/m² to 932.39 W/m², with average values of 4.73 °C for ambient temperature and 825.03 W/m² for solar intensity. The solar intensity steadily increased until noon and then began to decline, while the ambient temperature showed a continuous upward trend throughout the experiment. The figure also indicates that the performance evaluation of the three different PVTs was conducted under clear weather conditions.

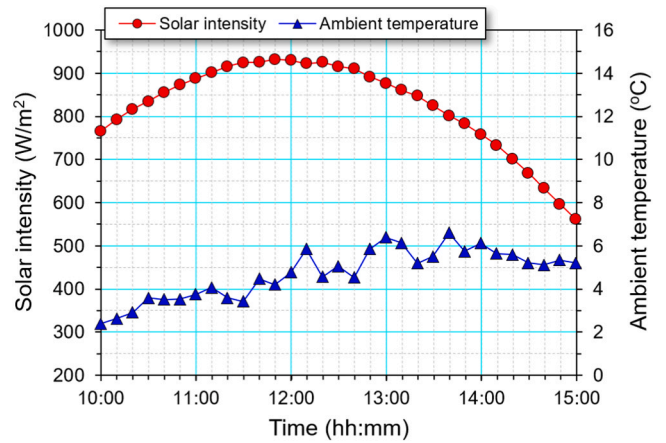


Fig. 6. Solar intensity and ambient temperature during the experiment.

3.2. Thermal performance

This subsection discusses the thermal performance of the PVTs, which was assessed to confirm the effectiveness of the heat transfer devices within the air channel of the proposed PVT. Fig. 7 illustrates the outlet air temperature for three different PVTs. The outlet air temperature rose with rising solar intensity and subsequently dropped as the solar intensity declined. During the experiment, the outlet air temperature varied from 5.32 °C to 9.99 °C for PVT-SMAC, from 7.04 °C to 11.85 °C for PVT-LF, and from 7.78 °C to 12.71 °C for PVT-LFRT. The average outlet air temperatures were 8.25 °C, 9.96 °C, and 10.76 °C for PVT-SMAC, PVT-LF, and PVT-LFRT, respectively. These results indicate that PVT-LFRT, which incorporates both longitudinal fins and rectangular turbulators, exhibited a higher outlet temperature compared with the other PVTs under identical weather conditions. This is because the turbulators installed on the fin surface can effectively boost the heat transfer between the fin surface and the flowing air.

Fig. 8 presents the heat gain of three different PVTs and their thermal efficiency recorded during the experiment. The heat gains of PVT-SMAC, PVT-LF, and PVT-LFRT varied within 113.9–247.84 W/

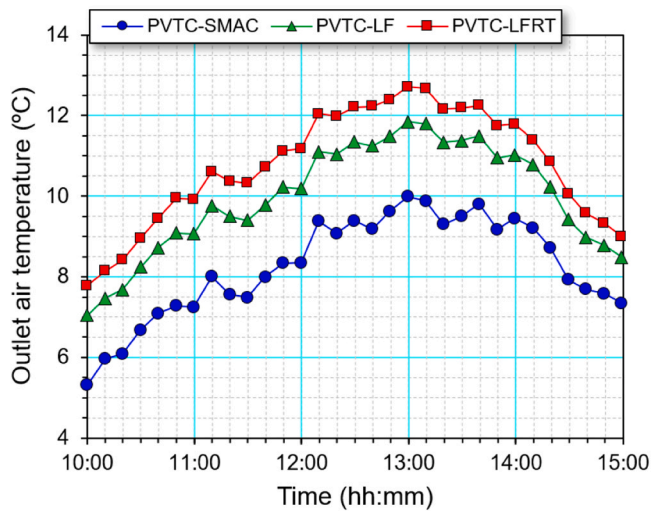


Fig. 7. Outlet air temperature of different PVTCs.

m^2 , 173.41–356.47 W/m^2 , and 202.08–410.39 W/m^2 , respectively, with average values of 187.65 W/m^2 , 277.41 W/m^2 , and 321.8 W/m^2 . This confirms that PVTC-LFRT, equipped with longitudinal fins and rectangular turbulators, provides a higher heat gain than PVTC-SMAC or

PVTC-LF does.

Additionally, the thermal efficiencies of PVTC-SMAC, PVTC-LF, and PVTC-LFRT ranged between 19.74–27.21 %, 29.45–39.13 %, and 34.32–45.05 %, respectively, with average values of 22.68 %, 33.56 %, and 38.93 %. The results indicate that PVTC-LFRT could achieve higher thermal efficiency than those of PVTC-SMAC and PVTC-LF during the experiment, under the same weather conditions.

The obtained results clearly show that installing rectangular turbulators on the fin surface effectively improves the heat gain of a PVTC and enhances its thermal conversion efficiency by facilitating the heat transfer between the fin surface and the flowing air.

3.3. Electrical performance

This subsection presents the results pertaining to the electrical performance of the PVTCs, confirming whether the different heat transfer devices installed in the PVTC affected its power generation and electrical efficiency. Fig. 9 plots the power generation of three different PVTCs and their electrical efficiency measured during the experiment. The power generation ranged within 93.77–154.36 W/m^2 for PVTC-SMAC, 94.43–155.24 W/m^2 for PVTC-LF, and 95.55–158.65 W/m^2 for PVTC-LFRT; the average values were 136.43 W/m^2 , 138.57 W/m^2 , and 140.75 W/m^2 , respectively. As solar intensity increases, power generation rises and decreases with decreasing solar intensity. The electrical efficiencies ranged within 16.02–16.96 % for PVTC-SMAC, 16.54–17.04 % for PVTC-LF, and 16.90–17.32 % for PVTC-LFRT, with

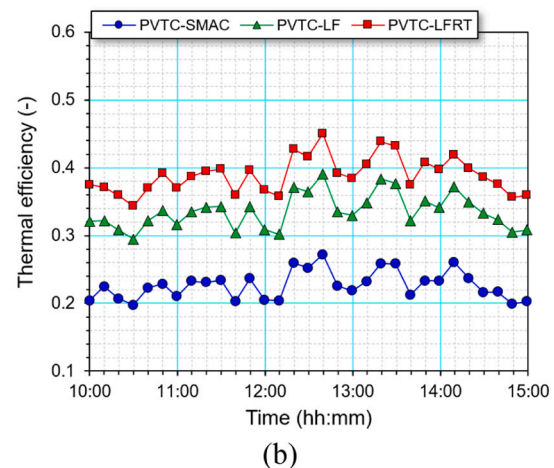
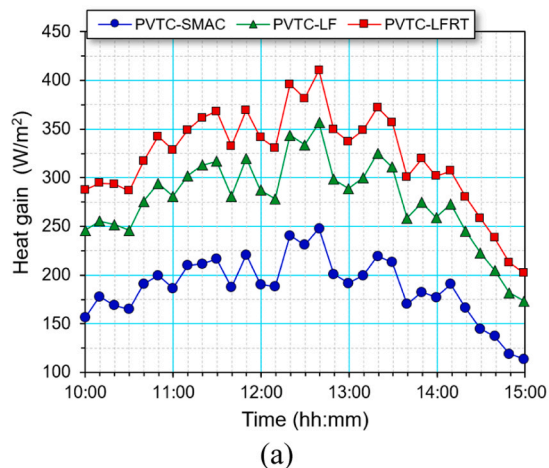


Fig. 8. (a) Heat gain and (b) thermal efficiency of different PVTCs.

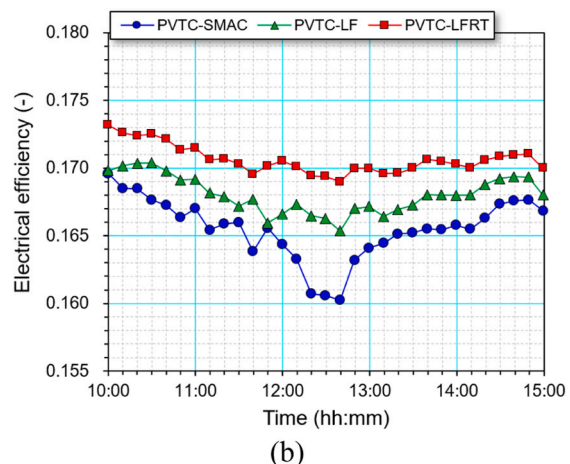
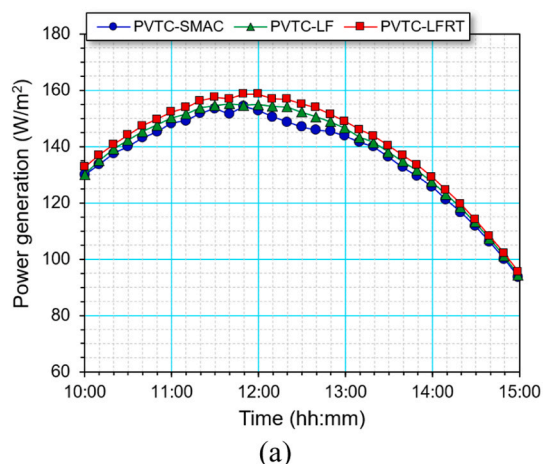


Fig. 9. (a) Power generation and (b) electrical efficiency of different PVTCs.

average values of 16.55 %, 16.80 %, and 17.06 %, respectively. The electrical efficiency declined slightly as the solar intensity increased but subsequently raised when the solar intensity dropped. These results demonstrate that PVTC-LFRT, which features both fins and rectangular turbulators, can achieve a higher average electrical efficiency during operation compared with PVTC-SMAC and PVTC-LF under the same weather conditions.

It is also observed that the PVTC-LFRT consistently exhibited higher power generation and electrical efficiency than PVTC-SMAC and PVTC-LF throughout the experiment. Generally, the cell temperature of the PV panel is inversely proportional to the electrical efficiency (Kim et al., 2021). In the case of PVTC-LFRT, the flowing air retrieved more thermal energy from the PV panel compared to the other types of PVTC due to the rectangular turbulators installed on the fin surface. Therefore, the cell temperature of the PV panel in PVTC-LFRT could be maintained at lower values than those of other types, enabling PVTC-LFRT to achieve a higher electrical efficiency. Consequently, it can be confirmed that adopting both rectangular turbulators and fins not only improves thermal performance but also yields higher electrical efficiency.

3.4. Thermal equivalent efficiency

This subsection presents the thermal equivalent efficiency, which considers both thermal and electrical efficiencies, for three different PVTCs. Fig. 10 shows the thermal equivalent efficiency during the experiment. For PVTC-SMAC, PVTC-LF, and PVTC-LFRT, the thermal equivalent efficiencies ranged between 63.37–69.59 %, 74.21–82.65 %, and 79.72–89.52 %, respectively. The average thermal equivalent efficiencies of PVTC-SMAC, PVTC-LF, and PVTC-LFRT were 66.23 %, 77.78 %, and 83.83 %, respectively. Compared to PVTC-SMAC, the addition of longitudinal fins in PVTC-LF improved the average thermal equivalent efficiency by 17.41 %, while the installation of both longitudinal fins and rectangular turbulators in PVTC-LFRT yielded a 26.57 % enhancement. Additionally, due to the presence of the rectangular turbulators, PVTC-LFRT achieved an approximately 6.07 % higher thermal equivalent efficiency than that of PVTC-LF.

Moreover, Table 6 compares the energy efficiency found in this work with similar PVTCs previously studied by other researchers. It indicates that the proposed PVTC could achieve better thermal equivalent efficiency than most previously studied PVTCs, as it not only enlarges the heat transfer area with fins but also improves the heat transfer coefficient with turbulators.

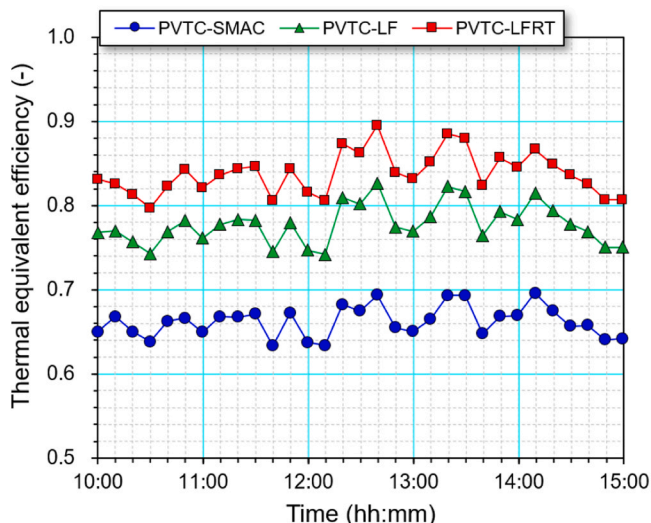


Fig. 10. Thermal equivalent efficiency of different PVTCs.

Table 6

Comparison of energy efficiency of air-based PVTC from previous studies.

Reference	Type of study	Efficiency (%)		
		Thermal	Electrical	Thermal equivalent
Solanki et al. (2009)	Experimental & Theoretical	42	8.4	64.11
Shahsavari and Ameri, (2010)	Experimental & Theoretical	50–60	9–10	73.68–86.32
Bambrook and Sproul, (2012)	Experimental	28–55	10.6–12.2	55.89–87.11
Ahn et al. (2015)	Experimental	23	15	62.47
Rounis et al. (2016)	Experimental	48	16.5	91.42
Mojumder et al. (2016)	Experimental	56.19	13.75	95.48
Tiwari et al. (2016)	Theoretical	14	11	42.95
Slimani et al. (2017)	Theoretical	44	10.65	72.03
Fan et al. (2017)	Theoretical	45–63	8–9	66.05–86.68
Huang et al. (2021)	Experimental	29.14–42.6	4.07–4.90	39.85–55.49
Wang et al. (2022)	Experimental	19.12–25.85	11.0–12.1	48.07–57.69
Kong et al. (2022)	Experimental	53.16	9.95	79.34
Cetina-Quinones et al. (2023)	Theoretical	23.34–27.92	13.16–13.43	57.97–63.26
Xie et al. (2024)	Experimental & Theoretical	37.2–42.3	11.3–12.5	66.94–75.19
Present study	Experimental	34.32–45.05	16.9–17.32	79.72–89.52

3.5. Daily energy output

This subsection compares the daily energy outputs of the three PVTCs obtained throughout the experiment to illustrate the impact of the heat transfer enhancement devices installed in the air channels. Fig. 11 presents the daily energy outputs of the PVTCs. The daily thermal outputs of PVTC-SMAC, PVTC-LF, and PVTC-LFRT were 0.97 kWh/m², 1.433 kWh/m², and 1.663 kWh/m², respectively. PVTC-LFRT achieved a significant increase of about 71.49 % and 16 % in daily thermal output with respect to PVTC-SMAC and PVTC-LF due to the presence of longitudinal fins and rectangular turbulators. Moreover, the daily electrical outputs were 0.705 kWh/m², 0.716 kWh/m², and 0.727 kWh/m² for PVTC-SMAC, PVTC-LF, and PVTC-LFRT, respectively. Daily power

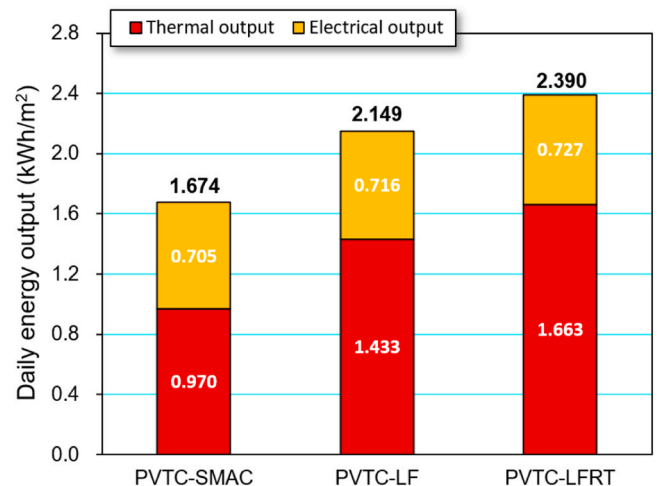


Fig. 11. Daily energy output of different PVTCs during the experiment.

generation during operation is improved by 3.17 % and 1.57 %, respectively, in PVTC-LFRT compared to PVTC-SMAC and PVTC-LF, as the installation of a longitudinal fin and rectangular turbulator boosted heat transfer between the PV panel and the flowing air, resulting in higher electrical efficiency.

Consequently, PVTC-LFRT achieved the highest daily energy output. The inclusion of longitudinal fins and rectangular turbulators improved its total daily energy output by approximately 42.73 % and 11.2 % over that of PVTC-SMAC and PVTC-LF, respectively.

4. Conclusion

In this work, a novel PVTC configuration incorporating both longitudinal fins and rectangular turbulators was proposed and evaluated experimentally under real-world weather conditions. The energy outputs and efficiencies of the suggested PVTC (PVTC-LFRT) were also compared with that of a PVTC equipped with a smooth air channel (PVTC-SMAC) and a PVTC equipped with only longitudinal fins (PVTC-LF) under identical weather conditions to verify that additional energy can be extracted from the incident solar energy through the proposed design. The conclusions obtained in this study are as follows:

- (i) Among the investigated PVTCs, PVTC combined with both longitudinal fins and rectangular turbulators (PVTC-LFRT) exhibited the highest outlet air temperature. During the experiment, the maximum outlet air temperature of PVTC-LFRT was observed to be 2.72 °C and 1.86 °C higher than those of PVTC with smooth air channel (PVTC-SMAC) and PVTC with only longitudinal fins (PVTC-LF), respectively.
- (ii) PVTC-LFRT consistently exhibited the highest thermal efficiency among the investigated PVTCs. During the experiment, the average thermal efficiency of PVTC-LFRT was observed to be 38.93 %, while those of PVTC-SMAC and PVTC-LF were found to be 22.68 % and 33.56 %, respectively.
- (iii) PVTC-LFRT achieved the highest electrical efficiency throughout the experiment. The average values of electrical efficiency were 17.06 % for PVTC-LFRT, 16.55 % for PVTC-SMAC, and 16.80 % for PVTC-LF.
- (iv) The thermal equivalent efficiencies of PVTC-SMAC, PVTC-LF, and PVTC-LFRT were 66.23 %, 77.76 %, and 83.83 %, respectively. Thus, the thermal equivalent efficiency of PVTC-LFRT was improved by 26.57 % and 6.07 % over those of PVTC-SMAC and PVTC-LF, respectively.
- (v) The inclusion of longitudinal fins and rectangular turbulators enhanced the total daily energy output of PVTC-LFRT by 42.73 % and 11.2 % compared with those of PVTC-SMAC and PVTC-LF, respectively.

The findings clearly indicate that attaching rectangular turbulators to longitudinal fins can effectively improve the performance of a PVTC in terms of both thermal and electrical efficiency, which confirms the feasibility of the proposed design. However, the performance of a PVTC with longitudinal fins and turbulators varies depending on the shape and installation conditions of the fins and turbulators. Therefore, future research should evaluate the PVTC's performance based on these factors and optimize both the shape and installation conditions. In addition, the main purpose of developing high-performance PVTCs is to minimize the energy consumption of conventional systems used for applications such as hot water supply, drying, and space heating. Hence, future studies will analyze the energy efficiency and economic aspects of conventional energy systems incorporating the proposed PVTC.

CRediT authorship contribution statement

Byung-Hwa An: Formal analysis, Data curation. **Kwang-Hwan Choi:** Writing – original draft, Investigation. **Seong-Bhin Kim:** Writing

– original draft, Visualization, Investigation, Formal analysis, Data curation. **Kwang-Am Moon:** Visualization, Data curation. **HwiUng Choi:** Writing – review & editing, Writing – original draft, Visualization, Investigation, Conceptualization.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data Availability

Data will be made available on request.

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